Data Transformation

When ANOVA assumptions fail: log, square root, angular and Box-Cox transformations, and back-transformation.

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1 Why transform?

Used when ANOVA assumptions fail, typically **unequal error variances (heteroscedasticity)**, non-normal errors or non-additive (multiplicative) effects.

1. Transform the data
2. Run ANOVA and mean separation on the **transformed** scale
3. **Back-transform** means to the original units for reporting

Never pick a transformation just to get the conclusion you want.

2 Which transformation?

Data	Variance	Use
Counts (Poisson)	$\sigma^2 = \mu$ or $c^2\mu$	**Square root**
Binomial percentages	$\sigma^2 = p(1-p)/n$	**Angular** (arcsine)
SD ∝ mean; multiplicative effects	$\sigma^2 = c^2\mu^2$	**Logarithmic**

3 Logarithmic

Positive data

$$X = \log Y$$

With zeros

$$X = \log(Y + 1)$$

Back-transform: $Y = \text{antilog}\, X$. Logs need positive inputs. Example in the book: weed counts.

4 Square root

Counts

$$X = \sqrt{Y}$$

Zeros present

$$X = \sqrt{Y + 1}$$

Very small counts: $\sqrt{Y} + \sqrt{Y+1}$. Back-transform $\sqrt{Y+1}$ with $Y = X^2 - 1$. Insect, larval, defective and diseased-seed counts.

5 Angular (arcsine) transformation

Transform

$$X = \sin^{-1}\sqrt{p}, \quad p = \frac{Y}{100}$$

0% and 100%

$$p = \frac{1}{4n}, \quad p = 1 - \frac{1}{4n}$$

Back-transform

$$Y = 100 \sin^2 X$$

For percentages from **binomial counts** (germination, mortality). The 30–70% guideline is approximate. Nutrient percentages (protein, carbohydrate) are not binomial and need not be transformed.

6 Box-Cox

A **family of power transformations** indexed by λ, for strictly positive data.

$\lambda = \frac{1}{2}$: square root · $\lambda = -1$: reciprocal · $\lambda \to 0$: log. The angular transformation is **not** in this family.

7 After transforming

- ✓ Check assumptions using the **residuals** of the fitted model.
- ✓ Use one transformation consistently for the whole dataset.
- ✓ CD on the transformed scale: $t_{\alpha/2,\,edf}\sqrt{2\,MSE/r}$ (RBD).
- ✓ Back-transformed means are **not** arithmetic means (e.g. squared root means are ≤ the original mean).

! Common mistakes

- × Reporting transformed means as final results.
- × Taking log of zero.
- × Arcsine on non-binomial percentages.

? Fill in the blanks

1. Counts of insects per plant are usually analysed after a _____ transformation. [square root]
2. Percent germination out of 100 seeds calls for the _____ transformation. [angular (arcsine)]
3. When the SD rises in proportion to the mean, use the _____ transformation. [logarithmic]
4. A transformed mean √(Y + 1) = 3 back-transforms to _____. [8]
5. An angular value of 30° back-transforms to _____ %. [25]
6. Counts that include zeros are transformed as √(Y + _____). [1]

Source: *Textbook of Agricultural Statistics*, Ch. 24 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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